

Effect of Graph Partitioning on Link Prediction with Bilinear, Hadamard, and Random Forest Decoders

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Motivation

- Large graphs that do not fit on one machine are embedded by partitioning the graph, embedding each partition independently, and merging the results.
- Each partition has its own skip-gram embedding space, so vectors from different partitions are not directly comparable.
- Link prediction asks whether embeddings generalize to edges that were **not** seen during training, which is what downstream recommendation depends on.
- Seminar 3 studied reconstruction only. Does a good reconstruction score imply good link prediction?

Problem Formulation

Given a static graph partitioned into P partitions, each embedded independently with static node2vec, how do

- the partition count P and the partitioning strategy (LFM vs. LPA),
- the link-prediction decoder (bilinear, Hadamard, random forest)

affect link-prediction quality, and how does that quality relate to reconstruction quality?

Partitioned Embedding Pipeline

- 1 Hold out 10% of the edges as positive test pairs; draw an equal number of non-edges as negatives.
- 2 Partition the training graph once into P partitions.
- 3 Embed each partition's induced subgraph independently with static node2vec.
- 4 Merge per-partition embeddings into one embedding per vertex; train the decoder on the merged embeddings.
- 5 Evaluate the decoder on the held-out pairs.

A pair whose endpoints lie in different partitions is scored from two independently trained spaces; this is expected to drive most of the loss.

Partitioners

Both are community-detection based: detect communities, then pack them into exactly P partitions with a size-balancing greedy bin-packing step.

- **LPA** (label propagation): communities are disjoint, so every vertex belongs to exactly one partition. No tunable parameters.
- **LFM** (Lancichinetti–Fortunato–Kertesz): greedy fitness-based growth from seed vertices; communities may overlap, so a vertex can appear in several partitions.

All main experiments use LPA.

Link-Prediction Decoders

Given merged embeddings $u = z(a)$, $v = z(b)$:

- **Hadamard:** $s(u, v) = w^\top (u \odot v) = \sum_i w_i u_i v_i$
- **Bilinear:** $s(u, v) = u^\top W v$, with $W \in \mathbb{R}^{d \times d}$ unrestricted
- **Random forest:** 20 trees on the concatenated vector $[u; v] \in \mathbb{R}^{2d}$, axis-aligned splits on raw coordinates

Neural decoders: $\hat{y} = \sigma(s(u, v))$, binary cross-entropy, minibatch SGD. An edge is predicted when $\hat{y} > 0.5$.

Datasets

Dataset	Nodes	Edges	Avg. deg.	Avg. clust.	Density	Modularity
CITESEER	3264	4536	2.78	0.145	0.0009	0.72
AstroPh	18772	198110	21.11	0.631	0.00112	0.33
AS-Oregon	11806	38781	6.57	0.399	0.00056	0.57
Cit-HepPh	34546	420921	24.37	0.285	0.00071	0.67

Four real-world graphs that vary in size, density, and community structure.

Embedding Parameters and Protocol

Dataset-specific node2vec parameters

Dataset	p	q	d
CITESEER	0.5	0.25	50
AstroPh	2	0.25	50
Cit-HepPh	4	0.25	100
AS-Oregon	0.5	2	128

- Fixed for all datasets: $r = 10$ walks, length $l = 80$, window $w = 10$, $K = 5$ negatives, 10 epochs.
- Ranking protocol: up to 1000 positive test edges, each ranked against 500 random negative candidates.
- $4 \times 4 \times 3 \times 10 = 480$ runs; each run repeats split, partitioning, embedding, and decoder training.
- Mann–Whitney U test and Cohen's d at $\alpha = 0.05$, no multiple-comparison correction.

Preliminary: LFM vs. LPA on CITESEER

Strategy	Partitions	LP accuracy	RESTORE F1
Baseline	1	0.9209 ± 0.0105	0.3782 ± 0.0110
LFM	2	0.6884 ± 0.0109	0.3154 ± 0.0122
LFM	4	0.5576 ± 0.0159	0.2693 ± 0.0087
LFM	8	0.5401 ± 0.0173	0.2099 ± 0.0065
LPA	2	0.8026 ± 0.0193	0.4516 ± 0.0360
LPA	4	0.7485 ± 0.0061	0.4816 ± 0.0128
LPA	8	0.7128 ± 0.0200	0.5401 ± 0.0107
LPA	16	0.6874 ± 0.0148	0.6264 ± 0.0133
LPA	32	0.6765 ± 0.0169	0.7017 ± 0.0049
LPA	64	0.6681 ± 0.0114	0.6922 ± 0.0047
LPA	128	0.6888 ± 0.0180	0.6569 ± 0.0141
LPA	256	0.6791 ± 0.0157	0.6293 ± 0.0199

Preliminary, single dataset, earlier protocol. LPA beats LFM at every shared partition count, which motivates LPA in the main experiment.

MRR (%) vs. Partition Count

Dataset	Decoder	$P=1$	$P=2$	$P=4$	$P=8$
CITESEER	Bilinear	27.61±1.82	17.90±1.80	13.80±0.71	10.94±1.17
	Hadamard	22.92±2.80	13.88±1.88	12.49±1.37	13.87±1.66
	RDF	32.27±1.47	28.02±2.03	24.75±1.77	23.99±1.37
AstroPh	Bilinear	68.74±1.32	51.03±1.55	45.19±2.19	39.64±2.28
	Hadamard	32.34±4.13	26.84±3.22	28.24±1.63	29.63±2.50
	RDF	62.73±1.28	62.05±1.87	62.70±1.40	61.95±0.91
Cit-HepPh	Bilinear	62.95±1.13	44.85±3.05	39.60±2.81	36.91±2.02
	Hadamard	37.05±2.79	28.38±1.22	27.15±1.43	32.15±1.98
	RDF	53.23±0.97	49.86±1.59	47.96±0.75	47.55±1.08
AS-Oregon	Bilinear	25.08±0.95	18.69±1.05	15.96±1.51	15.34±1.63
	Hadamard	14.48±1.68	10.47±0.78	9.12±1.08	9.97±0.86
	RDF	43.26±1.49	46.25±1.68	46.09±1.69	45.13±1.13

Mean $\pm$ std over 10 runs, LPA partitioning. Bold: best decoder per dataset and P .

Hits@1 (%) vs. Partition Count

Dataset	Decoder	$P=1$	$P=2$	$P=4$	$P=8$
CITESEER	Bilinear	17.70±1.14	11.08±1.79	8.08±0.99	5.76±1.10
	Hadamard	14.37±2.65	8.21±1.77	7.64±1.59	9.38±1.50
	RDF	23.91±1.43	21.55±1.96	18.79±1.72	18.28±1.61
AstroPh	Bilinear	56.79±1.58	38.54±1.70	32.57±2.29	27.68±2.34
	Hadamard	20.73±3.46	16.93±2.98	18.53±1.79	19.56±2.49
	RDF	54.44±1.42	53.86±1.85	54.79±1.53	53.81±1.38
Cit-HepPh	Bilinear	48.38±1.47	31.05±2.81	27.10±2.54	24.90±1.91
	Hadamard	22.45±2.49	17.08±1.24	16.08±1.42	19.93±1.69
	RDF	40.31±0.92	37.57±1.81	35.82±1.05	35.31±1.32
AS-Oregon	Bilinear	14.99±1.07	9.70±1.23	7.83±1.13	7.57±1.40
	Hadamard	6.99±1.44	4.92±0.73	4.08±0.88	4.49±0.91
	RDF	38.06±1.60	41.44±1.77	41.58±1.75	40.47±1.08

Mean $\pm$ std over 10 runs, LPA partitioning. Bold: best decoder per dataset and P .

Hits@3 (%) vs. Partition Count

Dataset	Decoder	$P=1$	$P=2$	$P=4$	$P=8$
CITESEER	Bilinear	32.25±2.93	19.69±1.80	15.54±1.02	12.01±1.69
	Hadamard	25.94±3.57	15.56±2.43	13.75±1.36	15.67±1.49
	RDF	35.70±2.07	30.40±2.22	26.91±2.05	26.14±1.59
AstroPh	Bilinear	77.48±1.50	58.30±2.02	51.70±2.66	44.71±3.24
	Hadamard	36.65±4.97	29.82±3.68	30.90±1.76	32.72±2.94
	RDF	66.61±1.61	66.12±2.10	66.49±1.64	65.79±0.88
Cit-HepPh	Bilinear	73.21±1.82	51.77±4.09	45.33±4.16	42.48±2.77
	Hadamard	43.07±3.61	32.01±1.33	30.70±1.90	37.74±2.51
	RDF	59.91±1.30	55.79±1.55	53.65±1.59	53.51±1.42
AS-Oregon	Bilinear	29.22±1.24	21.21±1.34	17.64±1.63	16.60±2.03
	Hadamard	15.45±2.20	10.43±1.09	9.12±1.22	10.19±1.17
	RDF	44.35±1.48	47.14±1.69	46.99±1.70	46.25±1.53

Mean $\pm$ std over 10 runs, LPA partitioning. Bold: best decoder per dataset and P .

Hits@10 (%) vs. Partition Count

Dataset	Decoder	$P=1$	$P=2$	$P=4$	$P=8$
CITESEER	Bilinear	47.37±3.50	30.71±2.55	24.50±1.34	20.73±1.73
	Hadamard	38.96±3.50	24.46±2.45	21.52±1.72	21.39±2.55
	RDF	48.23±2.61	39.67±2.87	35.59±2.35	34.02±2.01
AstroPh	Bilinear	89.91±0.78	74.46±1.37	69.70±1.91	63.97±2.55
	Hadamard	55.58±5.68	46.86±4.04	47.57±2.19	49.85±2.50
	RDF	78.96±1.45	77.90±2.41	77.85±1.23	77.67±0.97
Cit-HepPh	Bilinear	89.85±0.56	72.40±3.12	64.52±2.83	60.36±2.20
	Hadamard	67.77±3.84	52.46±1.58	50.80±1.67	57.23±2.52
	RDF	79.78±1.24	74.61±1.66	72.98±1.27	72.38±1.37
AS-Oregon	Bilinear	45.39±1.44	36.97±1.19	32.74±3.00	31.59±2.28
	Hadamard	29.62±2.72	21.46±1.93	18.50±2.07	20.43±1.33
	RDF	53.71±1.27	55.58±1.82	54.53±1.55	53.72±1.45

Mean $\pm$ std over 10 runs, LPA partitioning. Bold: best decoder per dataset and P .

Relative MRR Change from $P = 1$ to $P = 8$

Dataset	Bilinear	Hadamard	RDF
CITESEER	-60%	-39%	-26%
AstroPh	-42%	-8%	-1%
Cit-HepPh	-41%	-13%	-11%
AS-Oregon	-39%	-31%	+4%

- Bilinear loses 39–60% of its MRR.
- Hadamard loses 8–39%, mostly in the first step.
- Random forest changes between +4% and −26%.

Significance of Each Doubling of P (Cohen's d , MRR)

Dataset	Step	Bilinear	Hadamard	RDF
CITESEER	1 $\rightarrow$ 2	-5.35*	-3.79*	-2.40*
	2 $\rightarrow$ 4	-2.98*	-0.84	-1.72*
	4 $\rightarrow$ 8	-2.96*	+0.91*	-0.48
AstroPh	1 $\rightarrow$ 2	-12.27*	-1.49*	-0.43
	2 $\rightarrow$ 4	-3.07*	+0.55	+0.40
	4 $\rightarrow$ 8	-2.48*	+0.66	-0.64
Cit-HepPh	1 $\rightarrow$ 2	-7.86*	-4.03*	-2.55*
	2 $\rightarrow$ 4	-1.79*	-0.93	-1.52*
	4 $\rightarrow$ 8	-1.10*	+2.90*	-0.44
AS-Oregon	1 $\rightarrow$ 2	-6.37*	-3.06*	+1.88*
	2 $\rightarrow$ 4	-2.10*	-1.43*	-0.09
	4 $\rightarrow$ 8	-0.39	+0.86	-0.67

* Mann–Whitney $p < 0.05$, $n = 10$ per group.

Significance Summary

- **Bilinear**: 11 of 12 steps significant, all negative; the only non-significant step is AS-Oregon $4 \rightarrow 8$.
- **Hadamard**: 7 of 12 significant; $1 \rightarrow 2$ is negative everywhere, while two late steps are significantly positive (the MRR recovers).
- **Random forest**: 5 of 12 significant, confined to the first two doublings on CITESEER and Cit-HepPh, plus one positive step on AS-Oregon.

Partitioning has a large, monotone effect on bilinear, a mostly one-time effect on Hadamard, and a dataset-dependent, plateauing effect on the random forest.

Decoder Comparison (Cohen's d , MRR)

CITESEER and AstroPh

Dataset	P	B-H	R-B	R-H
CITESEER	1	+1.99*	+2.81*	+4.18*
	2	+2.18*	+5.28*	+7.23*
	4	+1.20	+8.11*	+7.75*
	8	-2.05*	+10.26*	+6.66*
AstroPh	1	+11.87*	-4.62*	+9.95*
	2	+9.57*	+6.42*	+13.38*
	4	+8.76*	+9.52*	+22.67*
	8	+4.18*	+12.83*	+17.19*

Cit-HepPh and AS-Oregon

Dataset	P	B-H	R-B	R-H
Cit-HepPh	1	+12.17*	-9.24*	+7.75*
	2	+7.08*	+2.06*	+15.12*
	4	+5.59*	+4.07*	+18.21*
	8	+2.38*	+6.57*	+9.67*
AS-Oregon	1	+7.77*	+14.52*	+18.14*
	2	+8.86*	+19.63*	+27.23*
	4	+5.21*	+18.78*	+26.02*
	8	+4.12*	+21.24*	+35.08*

B = bilinear, H = Hadamard, R = RDF; $d > 0$: first-named higher. * $p < 0.05$.

Decoder Ordering

- Most configurations are ordered **random forest** > **bilinear** > **Hadamard**.
- Random forest beats Hadamard significantly in all 16 groups on all four ranking metrics.
- Bilinear beats Hadamard significantly in 14 of 16 groups on MRR.
- Random forest beats bilinear on MRR in 14 of 16 groups; its advantage grows with P (AS-Oregon: d from +14.5 to +21.2).
- Exceptions: unpartitioned dense graphs. At $P = 1$, bilinear beats random forest on AstroPh (68.74% vs. 62.73%) and Cit-HepPh (62.95% vs. 53.23%).

RESTORE F1 (%) of the Same Embeddings

Dataset	$P=1$	$P=2$	$P=4$	$P=8$
CITeseer	35.88	43.92	49.72	55.68
AstroPh	70.62	67.76	67.84	67.75
Cit-HepPh	54.87	57.41	60.35	61.59
AS-Oregon	33.88	38.88	45.97	43.20

- F1 rises with P on CITESEER, Cit-HepPh, and AS-Oregon, while bilinear MRR falls by 60% on CITESEER.
- Partitioning makes local reconstruction easier; link prediction also needs cross-partition comparisons.

Reconstruction vs. Link Prediction

- Pooled Spearman ρ between per-run F1 and MRR is positive for all decoders ($\rho = 0.62\text{--}0.72$, $n = 160$).
- This is driven by dataset differences: graphs that are easy to embed score high on both.
- Within a dataset, increasing P pushes the two quantities apart.

Reconstruction F1 indicates how learnable a graph is, but is **not** a reliable proxy for how well partitioned embeddings support link prediction.

Conclusion

- Partitioning strongly degrades neural decoders: bilinear MRR falls by 39–60% and Hadamard by 8–39% from $P = 1$ to $P = 8$.
- Random forest on concatenated embeddings is far less sensitive (+4% to −26%).
- Decoder ordering: random forest > bilinear > Hadamard in most configurations; bilinear is best only on unpartitioned dense graphs.
- Reconstruction F1 rises with P while link prediction falls, so reconstruction is not a proxy for downstream link prediction.
- LPA beats LFM in a preliminary single-dataset comparison.

Future Work

- Split test pairs into within-partition and cross-partition pairs to test whether cross-partition pairs cause the degradation.
- Align per-partition embedding spaces (e.g. with a learned linear map) before decoding.
- Tune the neural decoders per partition count; repeat ranking evaluation with random tie-breaking.
- Extend the LFM vs. LPA comparison and the decoder study to more datasets.
- Study the replication factor as a way to recover link-prediction quality.

Thank you!